

GIUSEPPE FESTA

Control and AI Engineer for Humanoid Robots and Autonomous Systems

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Control Engineer with a passion for innovation and sustainability. I actively participated in patenting procedures and gained extensive leadership and product design experience managing startups, working for Michelin, Terna and Neura Robotics, and through the Alta Scuola Politecnica Program.

Currently managing a Tech startup in the field of Computer Vision and Robotics while developing control software for humanoid robots at Neura. Always seeking an opportunity to change the world.

PROFESSIONAL EXPERIENCE

- Neura Robotics | Zurich, Switzerland**
Embedded Control Engineer Team Lead for Humanoid Robots
2025-Present
- Design and deployment of RT systems on embedded platforms
 - Custom application development for robotic control systems and state estimations for navigation on humanoids
- Control Engineer for Humanoid Robots Metzingen, Germany 2025**
- Low-level control for actuators (Control synthesis and Firmware development).
 - Interface development for transparent command exchange via DDS systems
- Thinking! | Stage Robotics (startup) Milan, Italy - Lille, France**
Robotics Engineer
2022-2025
- Design of a custom robot, able to perform unique movements on the stage. The realisation includes 3D modelling and printing, Python and ROS2 software development, wiring, and RF coupling via Altium and Espressif IDE.
 - Design of Synchronised Systems and Firmware interfaces with DMX512 to control distributed systems during shows and live exhibitions, and production of documentation to support safety standards with Italian V.V.F.F. and French DCCRS.
- Michelin | Cuneo - Fossano Plant Cuneo, Italy**
Perception Engineer
2022-2024
- Successfully developing a fast and reliable fault detection system to improve online quality controls on the Michelin car tire production facility. The job required extensive Machine Learning skills, deep knowledge of OpenCV and TensorFlow libraries for C++ and Python, and optimization done through Vulcan for workload parallelisation.
- FreeRoom (freerom.digital) Milan, Italy - Valencia, Spain**
Co-Founder and CTO
2022-2024
- Founded a startup and implemented the perception and secure networking for meshed control cameras. The enterprise has since grown by acquiring partners interested in utilising advanced machine vision for security purposes.
- Dynamis PCR | PoliMi Reparto Corse Milan, Italy**
Inverter Designer
2020-2023
- In a team of three people, I successfully developed a three-phase 20 kW modular inverter to drive custom motors.
 - We developed the PCB and custom FOC software using MATLAB, OrCAD, CubeMX and STM32 Keil.

EDUCATION

- Politecnico di Milano Milan, Italy**
Msc. Automation and Control Engineering
2022-2024
- GPA:** 4.00 - 110 Cum Laude
 - Thesis:** Terna S.p.A.: *Design, Implementation and Experimental Tests of a Prototype of Power Converters for HVDC Applications.*
- Politecnico di Milano Milan, Italy**
Bsc. Automation and Control Engineering
2019-2022
- GPA:** 4.00 - 110
 - Relevant Activities:** Proven track record in Electronics, Machine Learning, Computer Vision, and Control Applications.
- Alta Scuola Politecnica Milan - Turin, Italy**
Member and Team Leader
2022-2024
- GPA:** 4.00
 - Completed the prestigious ASP honour program (PoliMi & PoliTo) focused on excellence in science and engineering. Participated in seminars with leading experts on technology, sustainability, and development. Completed research and management projects, including industry collaborations with Michelin, Terna, and other top enterprises.
- RWTH Aachen Aachen, Germany**
Erasmus+ Student
2023-2024
- FAU Erlangen - Siemens Additive Manufacturing Nuremberg, Germany**
Additive Manufacturing and Powders Processing Techniques Trainee.
2023-2024
- Università Parthenope di Napoli Naples, Italy**
CyberChallenge Cyber Security Training Program. Head of the Regional Team for the national phase.
2018 – 2020

ACTIVITIES & LEADERSHIP

- Athens Program Ambassador - Politechnika Warszawska Warsaw, Poland**
- Head of the team at the Warsaw University of Technology for the training program “Advanced Acoustic Measurements”.

SKILLS & INTERESTS

Interests: Boxing, MMA, Guitar, Rugby, Climbing, 3D Modelling, Aerial Photography

Languages: Italian (native), English (C2), Spanish (B2), German (A2), Turkish (A2)